

KATHMANDU UNIVERSITY
End Semester Examination [C]
January, 2018

Level : B.E./B. Sc./B. Tech

Course : MATH 104

Year : I

Semester : II

Exam Roll No. :

Time: 30 mins

F. M. : 20

Registration No. :

Date :

SECTION "A"
[10Q. $\times$ 1 = 10 marks]

Fill in the blank space(s) by writing the most appropriate word(s) or symbol(s).

1. The equivalent Cartesian equation of $1 = \cos\theta\sin\theta$ is
2. The gradient of $f(x, y, z) = x^2 + y^2 + z^2$ at point (1, 2, 3) is
3. The double integral $\iint \mathbf{R} r dr d\theta$ over $\mathbf{R}$ gives theof the closed and bounded region $\mathbf{R}$ in polar form.
4. If $1 - \frac{x^2 y^2}{3} < \frac{\tan^{-1} xy}{xy} < 1$ for all real values of x and y, $\lim_{(x,y) \rightarrow (0,0)} \frac{\tan^{-1} xy}{xy}$ is
5. $\left(\frac{\partial w}{\partial y}\right)_x$ is, if $w = x^2 + y^2 - z^2$, $z = x - y$.
6. The polar equation for the circle through the origin with radius 2 and center at (3, 0) is
7. $\int_0^2 \int_0^2 \int_0^2 dz dx dy = \dots\dots\dots$
8. The value of $\Gamma\left(\frac{1}{4}\right) \Gamma\left(\frac{3}{4}\right)$ is....., where $\Gamma(n)$ denotes the gamma function of n.
9. The fourier coefficient a_n of a function $f(x)$ of period 2π is defined by the formula $a_n = \dots\dots\dots$
10. The gradient field of $f(x,y,z) = xyz$ is.....

SECTION "B"
[10Q. $\times$ 1 = 10 marks]

Fill in the blank space(s), DO NOT TICK, by selecting the most appropriate answers from among the given ones.

11. The primitive period of $\tan nx$ is.....

$$\left[2\pi; \quad \pi; \quad \frac{\pi}{n}; \quad \frac{2\pi}{n} \right]$$

12. The curve $r = 5 + 5\cos \theta$ is symmetric about
 $[x - \text{axis}; \quad y - \text{axis}; \quad \text{origin}; \quad \text{line } y = x]$
13. $\beta(2, 3) = \dots\dots\dots$ where $\beta(m, n)$ denotes the beta function of m, n
 $\left[\frac{1}{10}; \quad \frac{1}{12}; \quad \frac{1}{14}; \quad \frac{1}{16} \right]$
14. The vector field $\vec{F}$ is conservative if and only if $\vec{F} = \dots\dots\dots$, where f is the scalar potential function.
 $[\nabla f; \quad \nabla F; \quad \nabla \times f; \quad \nabla \cdot f]$
15. The directrix of the parabola $r = \frac{25}{10 + 10\cos\theta}$ is
 $\left[5; \quad 2; \quad \frac{5}{2}; \quad \frac{2}{5} \right]$
16. The curvature of a circle with radius r is
 $\left[r; \quad 2r; \quad \frac{r}{2}; \quad \frac{1}{r} \right]$
17. If $f(x, y, z) = xy + z$ with $x = \cos t$, $y = \sin t$, $z = t$, then $\frac{df}{dt}$ at $t = 0$ is
 $[0; \quad 1; \quad 2; \quad 3]$
18. The plane containing unit tangent vector and binormal vector is called
 $[\text{principal plane}; \quad \text{rectifying plane}; \quad \text{osculating plane}; \quad \text{normal plane}]$
19. The polar equation $r^2 = \cos 2\theta$ is the
 $[\text{circle}; \quad \text{cardioid}; \quad \text{lemniscate}; \quad \text{four leaved rose}]$
20. If $f(x)$ is odd function, $\int_{-a}^a f(x) = \dots\dots\dots$
 $\left[0; \quad 2 \int_0^a f(x); \quad \int_a^{-a} f(x); \quad 2 \int_{-a}^a f(x) \right]$

KATHMANDU UNIVERSITY
End Semester Examination [C]
January, 2018

Level : B.E./B. Sc./B. Tech
Year : I
Time : 2 hrs. 30 mins.

Course : MATH 104
Semester : II
F. M. : 55

SECTION "C"

[4Q. $\times$ 7 = 28 marks]

1. Define level curve and level surface. Find the derivative of $f(x, y, z) = x^3 - xy^2 - z$ at $P_0(1, 1, 0)$ in direction of $\vec{v} = 2\hat{i} - 3\hat{j} + 6\hat{k}$. In what direction f change most rapidly at point P_0 and what are the rates of change in these directions? [1+1+ 5]
2. Define Beta and Gamma Function. Prove that:
$$\beta(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$$
 [1 + 1 + 5]
3. What is the path independence? Write the partial derivative criteria for a conservative vector field $\vec{F}$. Show that $\vec{F} = (e^x \cos y + yz)\hat{i} + (xz - e^x \sin y)\hat{j} + (xy + z)\hat{k}$ is conservative and find a potential function for $\vec{F}$. [1+1+2+3]

OR

Define unit tangent vector and binormal vector. Find $\vec{v}, \vec{T}, \vec{N}$, and $\vec{B}$ at the given value of t :

$\vec{r}(t) = (\cos t)\vec{i} + (\sin t)\vec{j} - \vec{k}$ at $t = \pi/4$ (where the symbols have their usual meaning)

[2+1+1+1.5+1.5]

4. State the normal and tangential form of Green's theorem. Verify normal form of Green's theorem for the field: $F(x, y) = (x - y)\hat{i} + x\hat{j}$ and the region R bounded by the unit circle C : $\vec{r}(t) = (\cos t)\hat{i} + (\sin t)\hat{j}$, $0 \leq t \leq 2\pi$. [1+1+5]

SECTION "D"

[9Q. $\times$ 3 = 27 marks]

5. Find the local extreme value of the function: $f(x, y) = xy - x^2 - y^2 - 2x - 2y + 4$
6. Find the average value of $F(x, y, z) = x^2 + 9$ over the cube in the first octant bounded by the coordinate planes and the planes $x = 2$, $y = 2$, and $z = 2$.
7. Sketch the graph and then find the area of the region $r = 1 - \cos \theta$.
8. Define first partial derivative of the function $f(x, y)$ with respect to y and find $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ at the point $(1, 2)$ if $f(x, y) = x^2 + 3xy - y^2$.

9. Find the spherical and cylindrical coordinates of the point whose Cartesian coordinate is $(0, 1, 0)$.
10. Using double integral find the area of the region **R** bounded by $y = 2x^2$ and $y^2 = 4x$.

OR

Find the volume of the region that lies under the paraboloid $z = x^2 + y^2$ and above the triangle enclosed by the lines $y = x$, $x = 0$ and $x + y = 2$ in the xy - plane.

11. Find the Fourier sine series of the function: $f(x) = x$, for $0 < x < \pi$.
12. What is the line integral? Find the line integral of $f(x, y, z) = x - 3y^2 + z + 5$ over the straight line segment $x = y = z = t$, $0 \leq t \leq 1$.
13. Verify Stoke's theorem for the hemisphere $S: x^2 + y^2 + z^2 = 25, z \geq 0$ its bounding circle $C: x^2 + y^2 = 25, z = 0$ and field $\vec{F} = y\hat{i} - x\hat{j}$.